



National Vocational and Technical Training Commission

SUBMITTED BY: Suleman Ahmad

COURSE TITLE: NAVTTTC AI

ROLL NO: CND-02325528

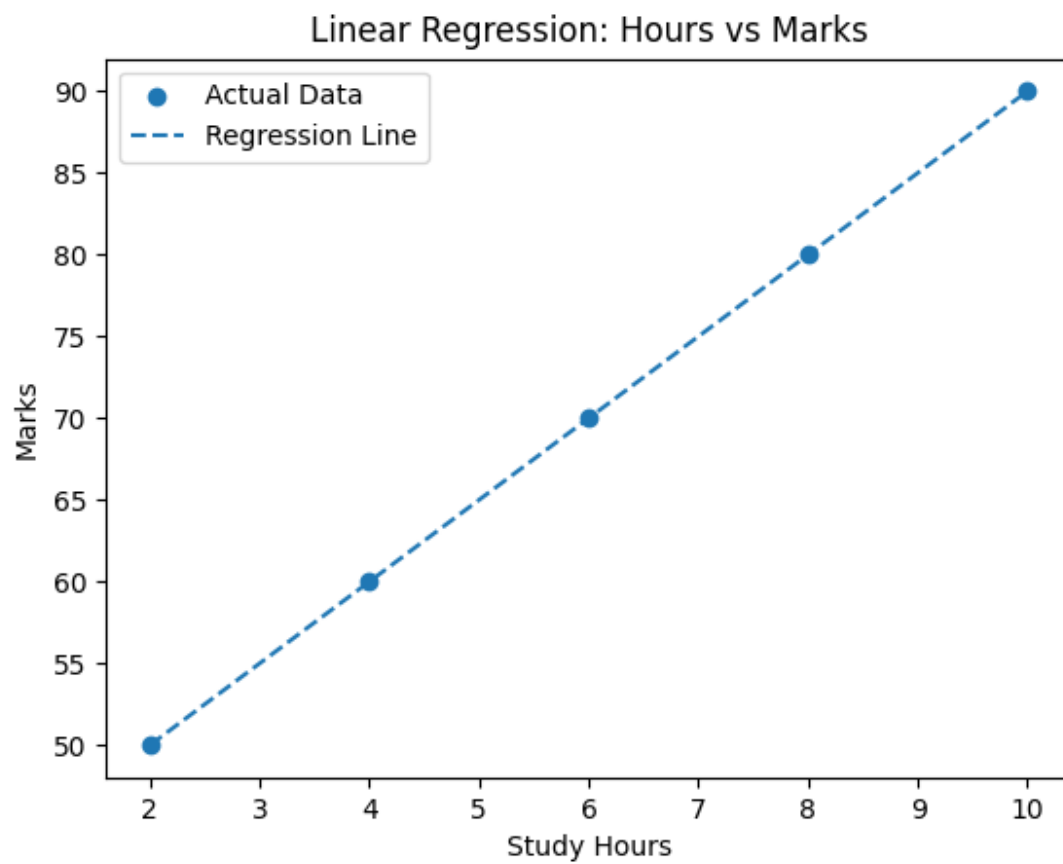
TASK: Week#5

Task_51

```
51.py > ...
1  import numpy as np
2  import matplotlib.pyplot as plt
3  from sklearn.linear_model import LinearRegression
4
5  # Simple Dataset
6  hours = np.array([2, 4, 6, 8, 10]).reshape(-1, 1)
7  scores = np.array([50, 60, 70, 80, 90])
8
9  # Create Model
10 model = LinearRegression()
11 model.fit(hours, scores)
12
13 # Prediction line
14 predicted_scores = model.predict(hours)
15
16 # ===== ONE IMPORTANT GRAPH =====
17 plt.scatter(hours, scores, label="Actual Data")
18 plt.plot(hours, predicted_scores, linestyle='--', label="Regression Line")
19
20 plt.xlabel("Study Hours")
21 plt.ylabel("Marks")
22 plt.title("Linear Regression: Hours vs Marks")
23 plt.legend()
24 plt.show()
25
26 # Output
27 print("Slope:", model.coef_[0])
28 print("Intercept:", model.intercept_)
29
30 # Prediction example
31 print("Marks for 7 hours:", model.predict([[7]])[0])
```

OUTPUT

```
Predicted score for 7 study hours: 75.0
Slope: 4.999999999999999
Intercept: 40.00000000000001
Predicted marks for 7 hours: 75.0
```



Task_52

```
52.py > ...
1  import numpy as np
2  import matplotlib.pyplot as plt
3  from sklearn.linear_model import LogisticRegression
4
5  # Data (Hours vs Pass/Fail)
6  X = np.array([1, 2, 3, 4, 5, 6, 7, 8]).reshape(-1, 1)
7  y = np.array([0, 0, 0, 0, 1, 1, 1, 1]) # 0 = Fail, 1 = Pass
8
9  # Model
10 model = LogisticRegression()
11 model.fit(X, y)
12
13 # Prediction curve
14 x_line = np.linspace(1, 8, 100).reshape(-1, 1)
15 y_prob = model.predict_proba(x_line)[:, 1] # Probability of class 1 (Pass)
16
17 # ===== ONE GRAPH =====
18 plt.scatter(X, y, label="Actual Data")
19 plt.plot(x_line, y_prob, linestyle='--', label="Logistic Curve")
20
21 plt.xlabel("Study Hours")
22 plt.ylabel("Probability of Passing")
23 plt.title("Logistic Regression: Pass/Fail Prediction")
24 plt.legend()
25 plt.show()
26
27 # Prediction
28 print("Probability of passing for 5 hours:", model.predict_proba([[5]])[0][1])
29 print("Class prediction (0=Fail,1=Pass):", model.predict([[5]])[0])
```

```
[Running] python -u "c:\Users\B-T\Desktop\ALLaboutAI\week5\week5_task\52.py"
Probability of passing for 5 hours: 0.6421915702717961
Class prediction (0=Fail,1=Pass): 1
```

